1.17 (1) $\bar{F}=(\bar{A}+\bar{B})(A\bar{B}+\bar{C}\bar{D}\bar{E})$

$F'=(A+B)(\bar{A}B+CDE)$

(3) $\bar{F}=\bar{A}\odot B\odot 0=\bar{A}\odot\bar{B}$

$F'=A\odot\bar{B}\odot 0=A\odot B$

1.18 证明：

(1) $AB+BC+CA=(A+B)(B+C)(C+A)$

右式$=(AB+AC+B+BC)(C+A)$

$=ABC+AC+BC+AB$

$=AB+BC+CA$

(2) $(X\oplus Y)\oplus Z=X\oplus(Y\oplus Z)$

右式$=X\oplus(Y\bar{Z}+\bar{Y}Z)$

$=\bar{X}(Y\bar{Z}+\bar{Y}Z)+X(\bar{Y}+Z)(Y+\bar{Z})$

$=\bar{X}Y\bar{Z}+\bar{X}\bar{Y}Z+X\bar{Y}\bar{Z}+XYZ$

左式$=(X\bar{Y}+\bar{X}Y)\oplus Z$

$=(X\bar{Y}+\bar{X}Y)\bar{Z}+(\bar{X}+YX+\bar{Y})Z$

$=X\bar{Y}\bar{Z}+\bar{X}Y\bar{Z}+\bar{X}\bar{Y}Z+XYZ$

∴左式=右式 即 $(X\oplus Y)\oplus Z=X\oplus(Y\oplus Z)$

1.20 (1) $F = \overline{(AB + ABD)(B + CD)}$

$= \overline{AB}(B + CD)$

$= (\bar{A} + \bar{B})(B + CD)$

$= \bar{A}B + \bar{A}CD + \bar{B}CD$

$= \bar{A}B(C + \bar{C})(D + \bar{D}) + \bar{A}(B + \bar{B})CD + (A + \bar{A})\bar{B}CD$

$= \bar{A}BCD + \bar{A}BC\bar{D} + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} + \bar{A}\bar{B}CD + A\bar{B}CD$

$= m_7 + m_6 + m_5 + m_4 + m_3 + m_{11}$

$= \Sigma m^4(3, 4, 5, 6, 7, 11)$

$= \Pi M^4(0, 1, 2, 8, 9, 10, 12, 13, 14, 15)$

(2) $F = (\bar{A} + C)(A + B)(C + \bar{D})$

$= (\bar{A}B + AC + BC)(C + \bar{D})$

$= \bar{A}BC + \bar{A}B\bar{D} + AC + AC\bar{D} + BC + BC\bar{D}$

$= \bar{A}BC + \bar{A}B\bar{D} + AC + BC$

$= \Sigma m^4(4, 6, 7, 10, 11, 14, 15)$

$= \Pi M^4(0, 1, 2, 3, 5, 8, 9, 12, 13)$

D \ AB	00	01	11	10
00	0	4 ✓	12	8
01	1	5	13	9
11	3	7 ✓	15 ✓	11 ✓
10	2	6 ✓	14 ✓	10 ✓

C

(3) $F = (\bar{A} \oplus B)(A \oplus \bar{B}) + B \oplus C \oplus D$

$= \bar{A} \oplus B + B \oplus C \oplus D$

$= \bar{A}\bar{B} + AB + \bar{B}\bar{C}D + B\bar{C}\bar{D} + \bar{B}C\bar{D} + BCD$

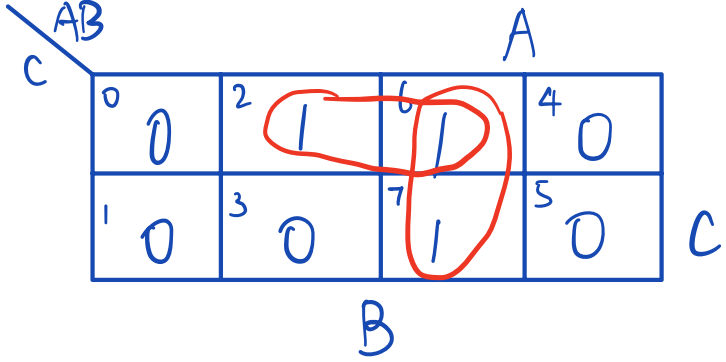
$= \Sigma m^4(0, 1, 2, 3, 4, 7, 9, 10, 12, 13, 14, 15)$

$C = \Pi M^4(5, 6, 8, 11)$

D \ AB	00	01	11	10
00	0 ✓	4 ✓	12 ✓	8
01	1 ✓	5	13 ✓	9 ✓
11	3 ✓	7 ✓	15 ✓	11
10	2 ✓	6	14 ✓	10 ✓

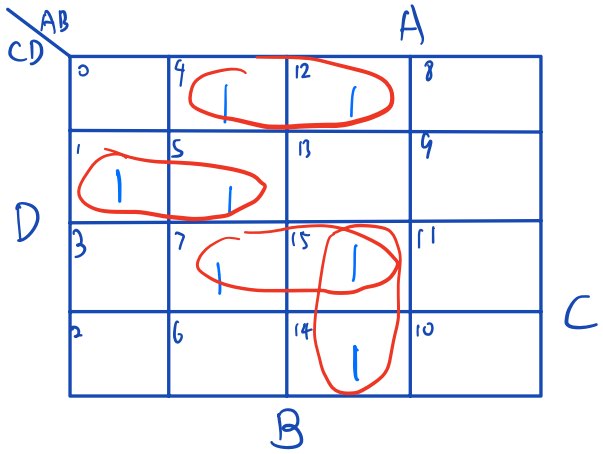
B

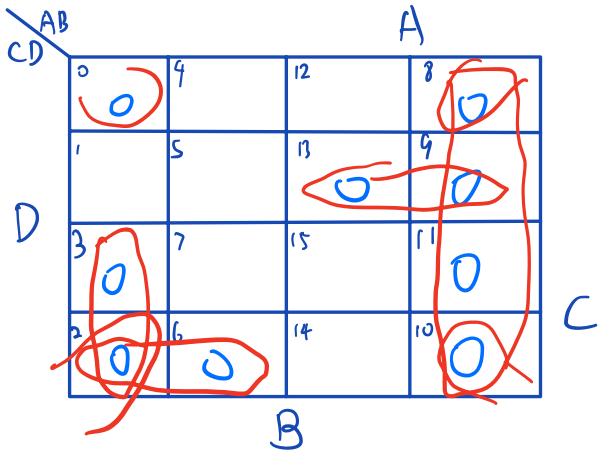
1.22 (2)



$$F = B\bar{C} + AB = B(A + \bar{C})$$

(3)



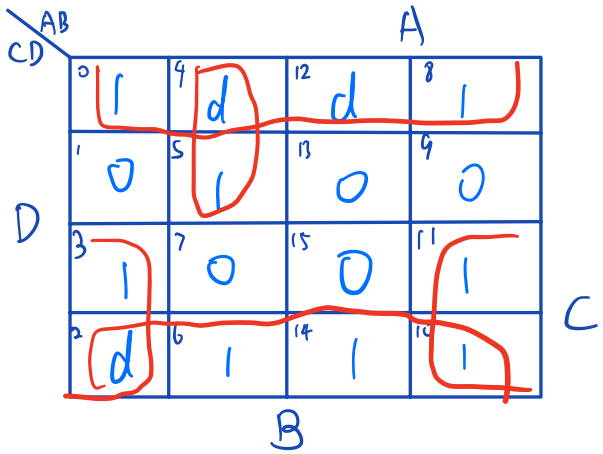


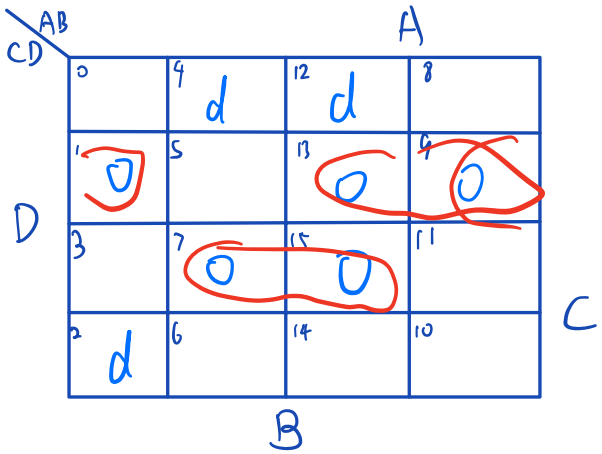
① $F = B\bar{C}\bar{D} + \bar{A}\bar{C}D + BCD + ABC$

② $\bar{F} = A\bar{B} + \bar{B}\bar{D} + A\bar{C}D + \bar{A}\bar{B}C + \bar{A}C\bar{D}$

$F = (\bar{A} + B)(B + D)(\bar{A} + C + \bar{D})(A + B + \bar{C})(A + \bar{C} + D)$

(4)

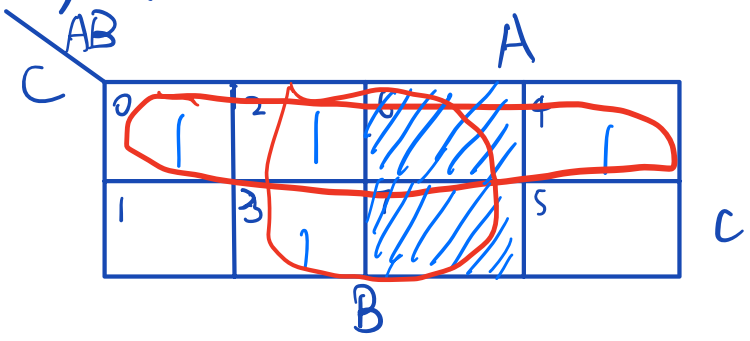




① $F = \bar{D} + \bar{B}C + \bar{A}B\bar{C}$

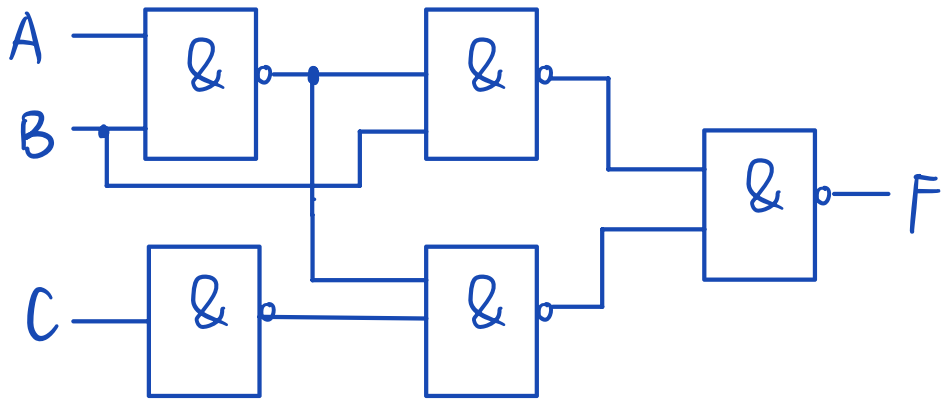
② $\bar{F} = BCD + A\bar{C}D + \bar{B}\bar{C}D \quad \therefore F = (\bar{B} + \bar{C} + \bar{D})(\bar{A} + C + \bar{D})(B + C + \bar{D})$

1.23 (3) $F = \bar{A}\bar{C} + \bar{A}B + B\bar{C} = \bar{A}B + B\bar{C}$



$$F = \overline{\bar{C}\,\overline{AB} + B\overline{AB}}$$

$$= \overline{\bar{C}\,\overline{AB}}\;\overline{B\overline{AB}}$$



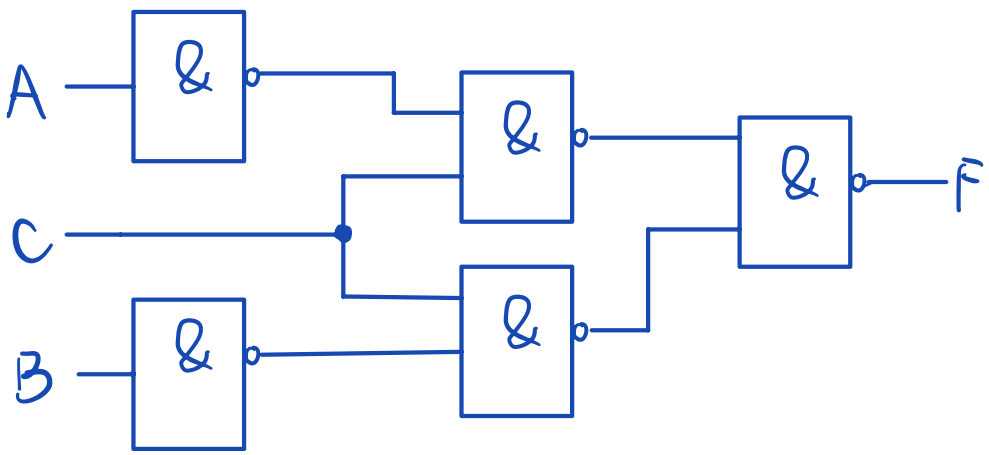
(4) $F = (\bar{A} + \bar{B})(AB + C)$

$$= \bar{A}C + \bar{B}C$$

$$= \overline{\bar{A}C}\;\overline{\bar{B}C}$$

$$\bar{F} = AB + \overline{AB}\cdot\bar{C}$$

$$F = \overline{\overline{AB}\cdot C}$$



1.24 解：$F = WX \oplus YZ$

$F = \bar{A}BC + A\bar{B}\bar{C}$

$= B(\bar{A}C + A\bar{C})$

$= B(A \oplus C)$

$= AB \oplus BC$

∴ $WX = AB$，$YZ = BC$

∴ 可能情况：$W = A$，$X = Y = B$，$Z = C$

1.25 解：扑克牌：四种花色 A～K（13张）+大小王

花色：红桃(00)，黑桃(01)，方片(10)，梅花(11)

点数：A(0001)，2(0010)，3(0011)，4(0100)，

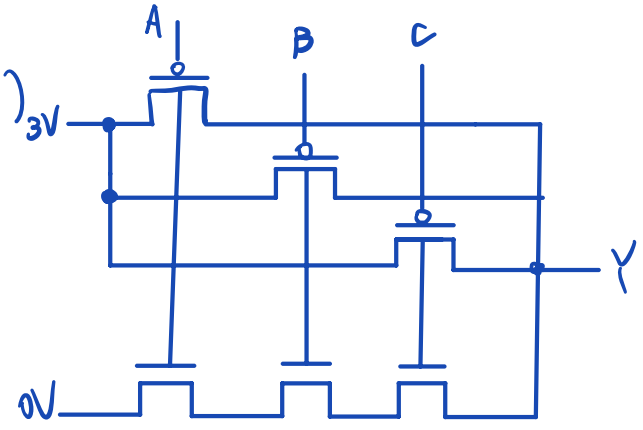
5(0101)，6(0110)，7(0111)，8(1000)，

9(1001)，10(1010)，J(1011)，Q(1100)，

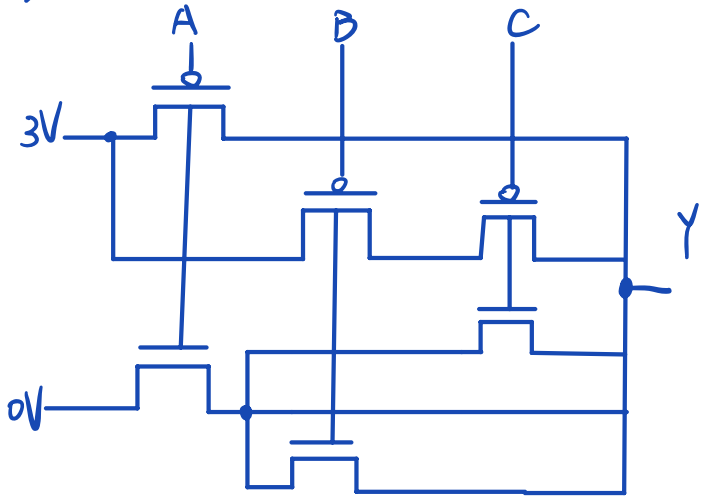
K(1101)，小王(1110)，大王(1111)

例：黑桃9(01 1001)，方片3(10 0011)，大王(11 1111)

1.26 (1)



(3) $Y = \overline{A + BC} = \bar{A}(\bar{B} + \bar{C})$



1.28 (1)有极性输出：输出信号有固定极性，始终保持在一个固定的电平。

(2)集电极开路门：

①实现线与功能：多个OC门直接并联，通过公共上拉电阻实现"逻辑与"。

②电平转换：上拉电阻连接到不同电压源，实现不同电压系统接口转换。

③驱动负载：驱动LED、继电器等高电流负载。

(3)三态门：

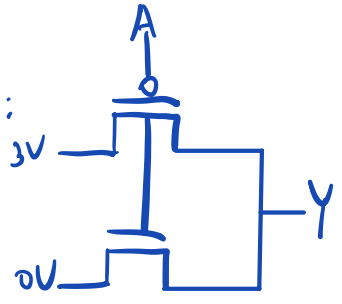
①高阻态：允许多设备共享同一总线而互不干扰。

②总线共享：多个三态门分时控制总线，减少连线数量。

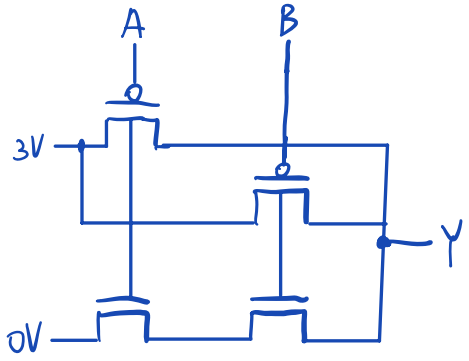
③动态控制：通过使能信号灵活控制数据传输方向。

1.29 原理：CMOS传输门由PMOS和NMOS管并联构成，栅极由互补信号C和$\overline{C}$控制。当C=1，$\overline{C}$=0时，PMOS与NMOS导通，输入信号双向传输至输出；当C=0，$\overline{C}$=1时，两管均截止，输出高阻态。

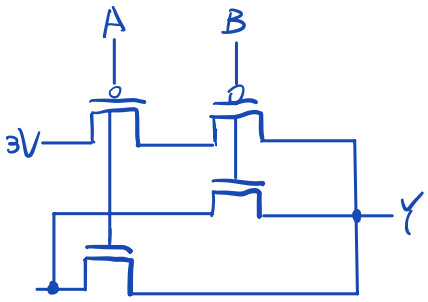
非门：



与非门：



或非门：



1.30 工艺：CMOS电路和TTL电路

规模：小规模、中规模、大规模、超大规模、巨大规模

特性：负载能力、延迟特性、功耗特性、管脚处理